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Nurenberg et al.

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(54) GABLE VENT ASSEMBLY WITH OFFSET LOCK

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F24F 7/02 (2006.01)

(52) **U.S. Cl.**
CPC ***F24F 7/02*** (2013.01)

(58) **Field of Classification Search**
CPC F24F 7/02; F24F 13/082
See application file for complete search history.

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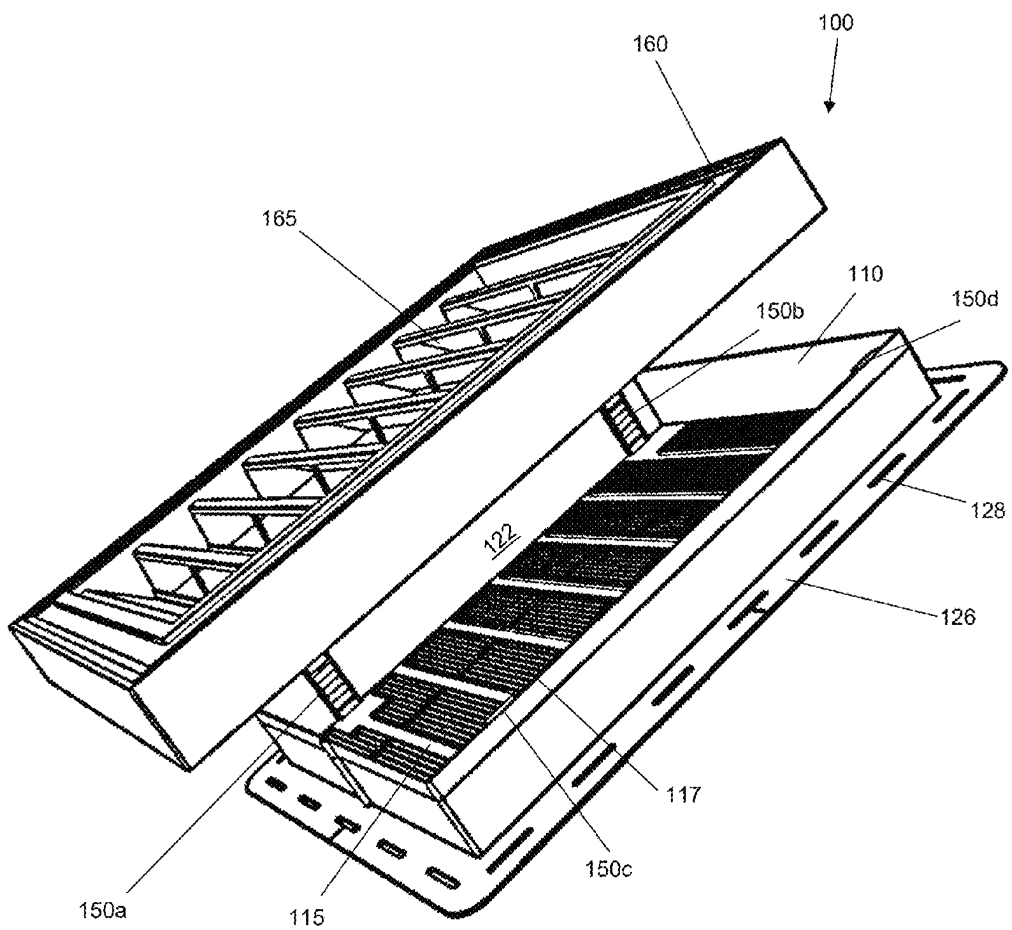
(57) **ABSTRACT**

A gable vent assembly has a base that mates and locks with a cover having vents in a first position when locks of the base and cover are aligned with one another to lock together. The cover and base nest together in a second unlocked position when the cover is rotated relative to the base so that locks of the base and cover are not aligned and cannot lock together.

8 Claims, 12 Drawing Sheets



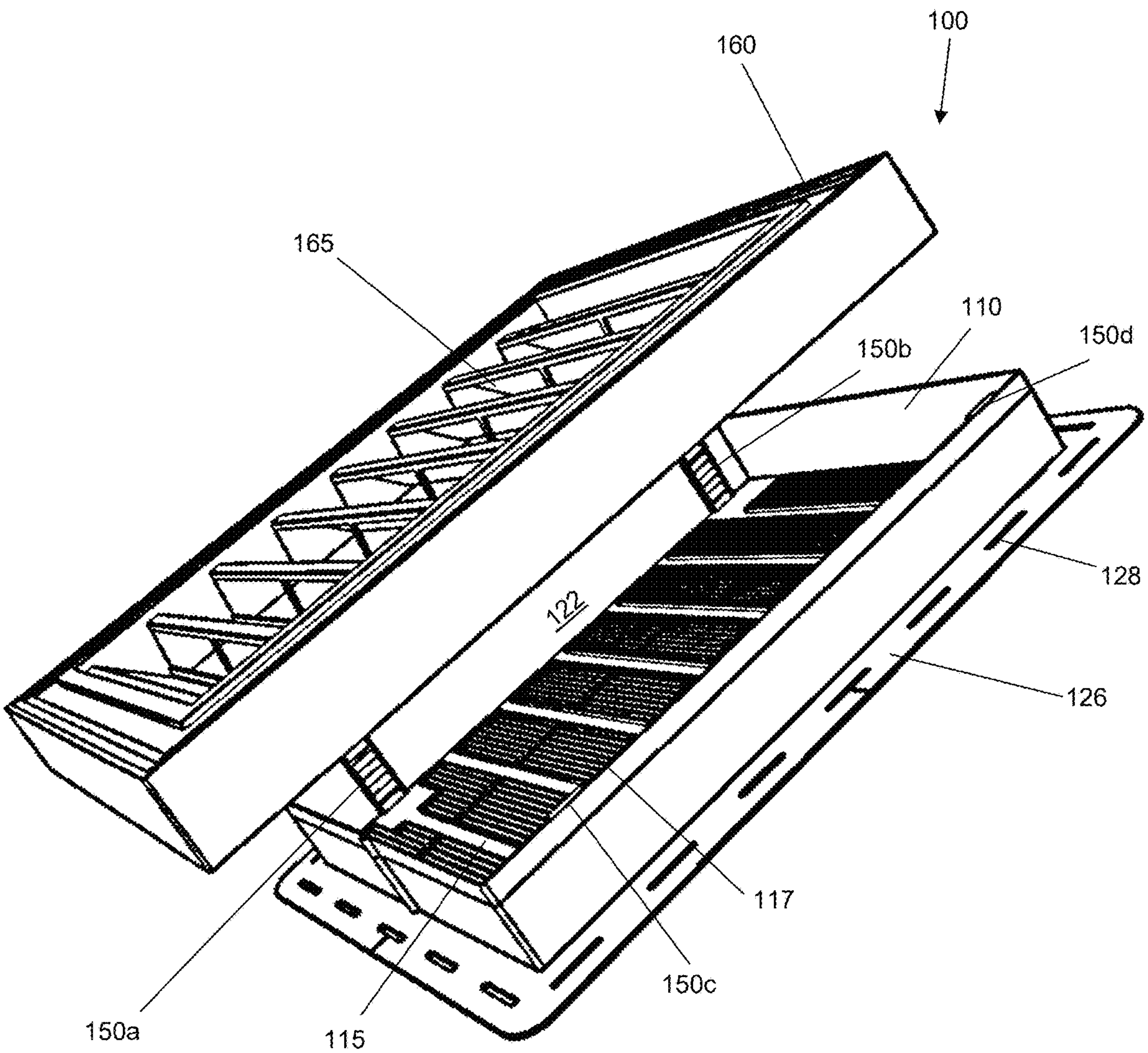


FIG. 1

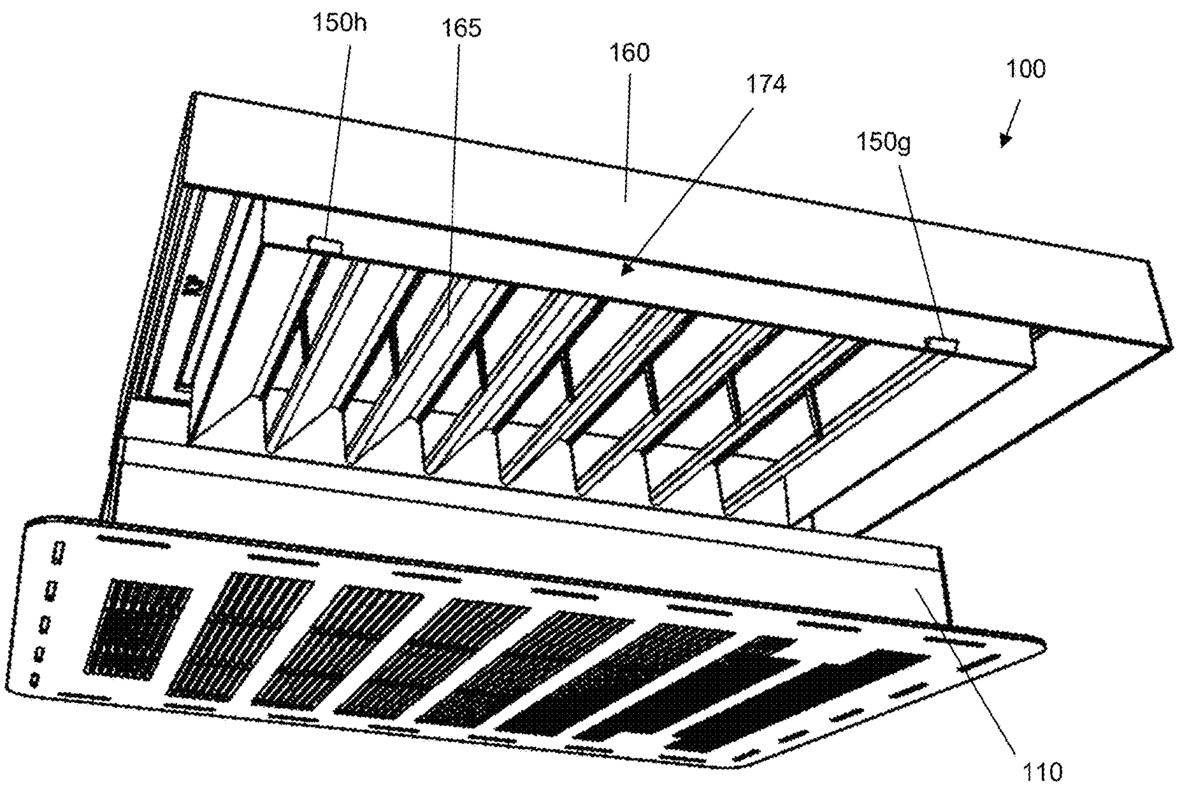


FIG. 2

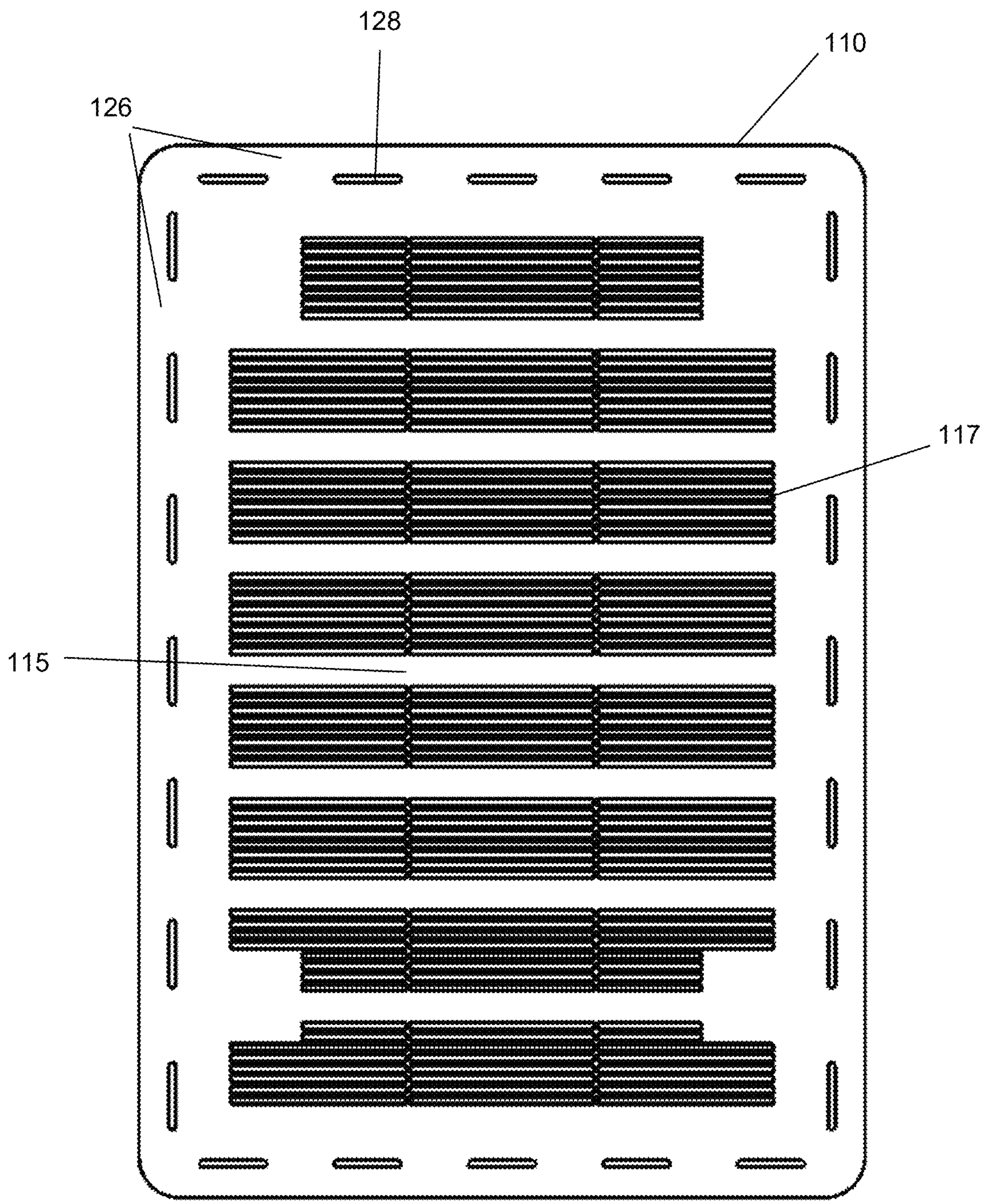


FIG. 3

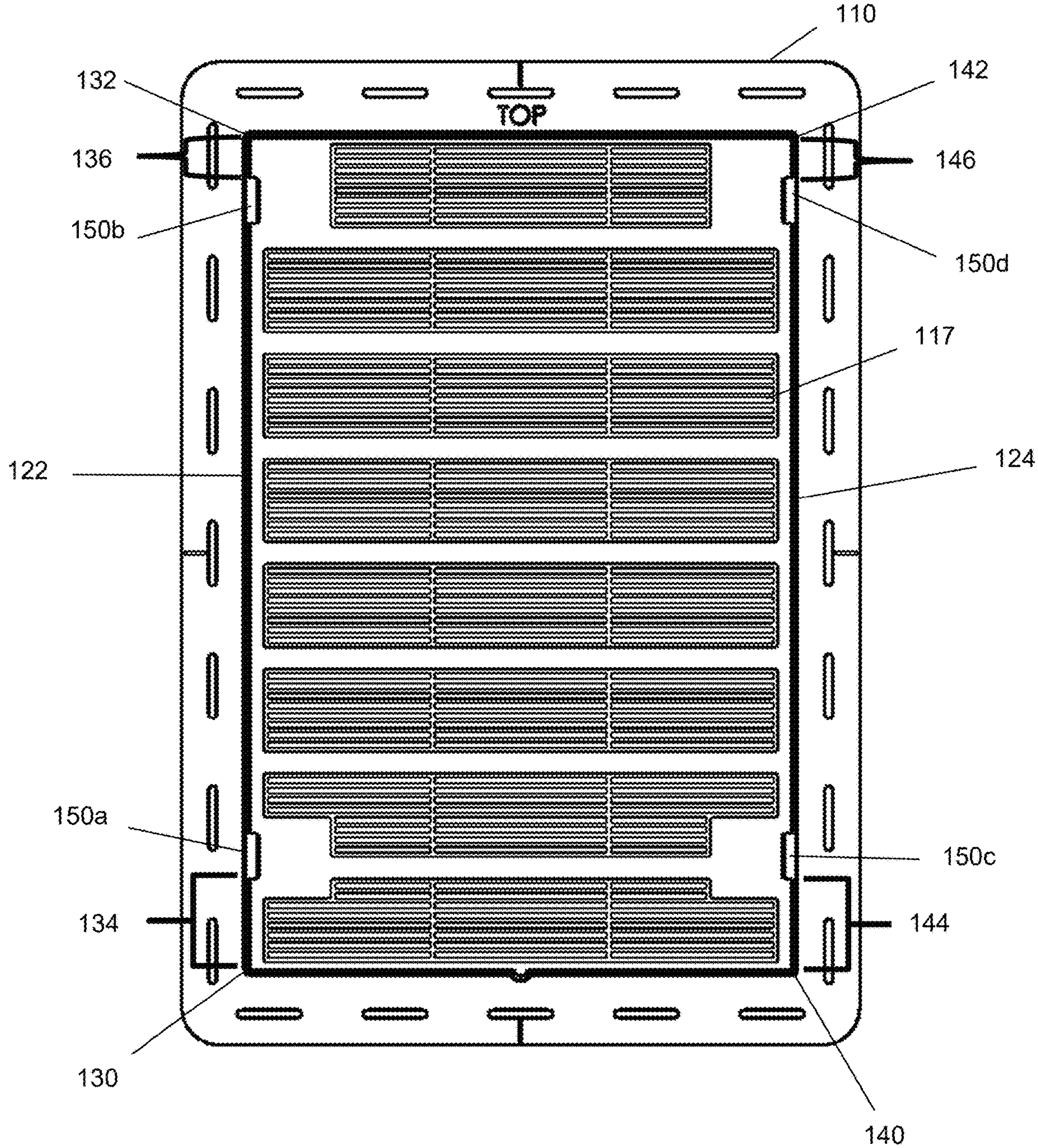

110
132
142
TOP
136
146
150b
150d
117
122
124
150a
150c
134
144
130
140

FIG. 4

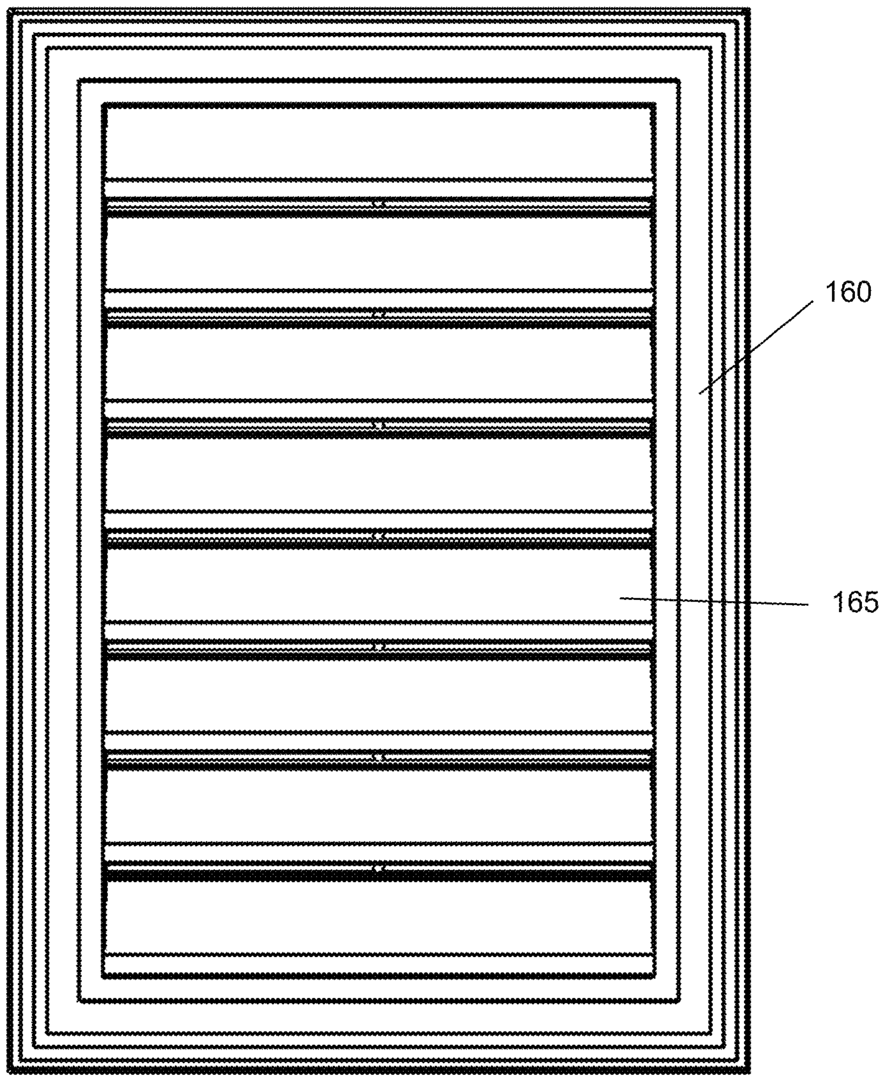


FIG. 5

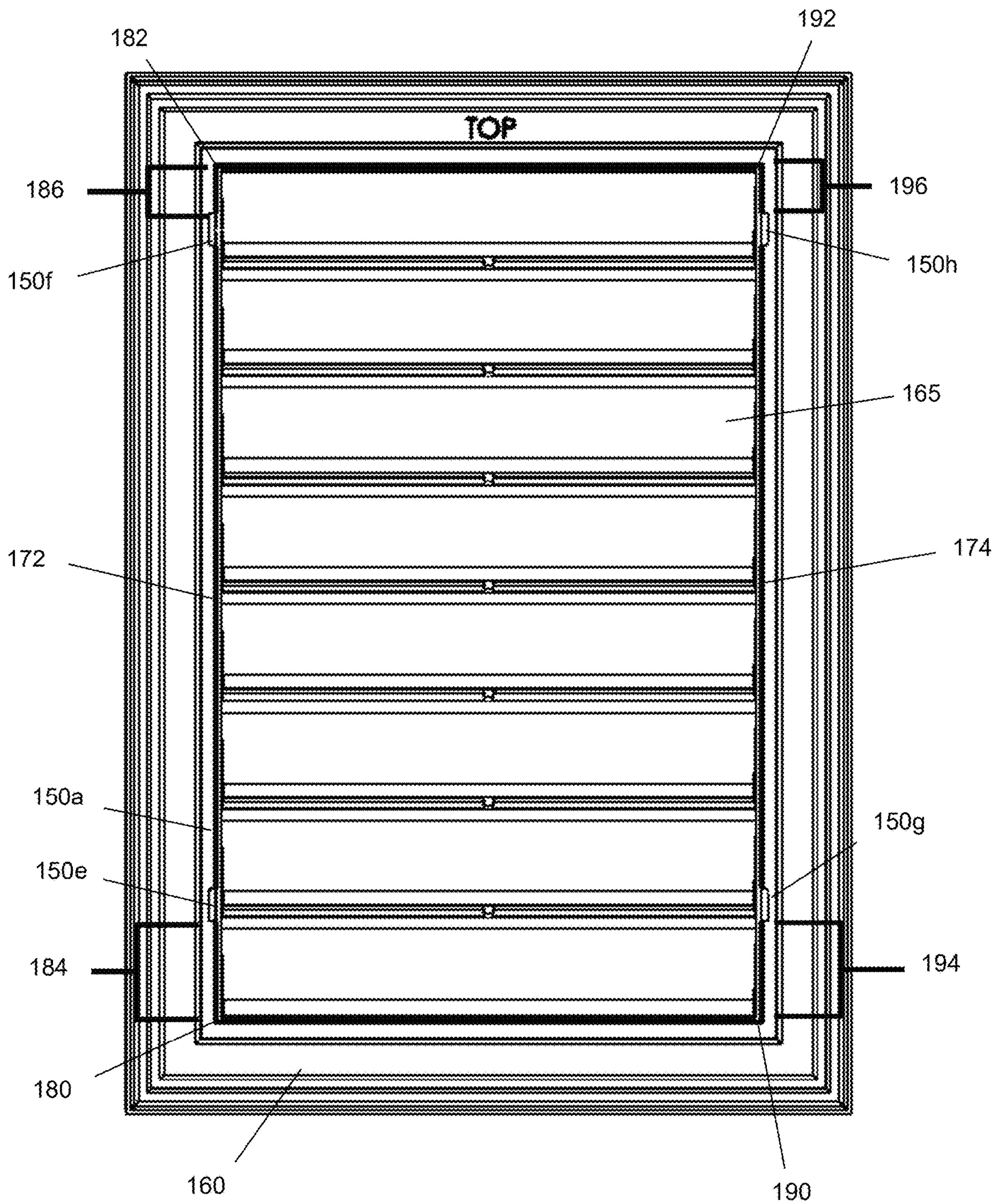

182
192
TOP
186
196
150f
150h
165
172
174
150a
150g
150e
184
194
180
160
190

FIG. 6

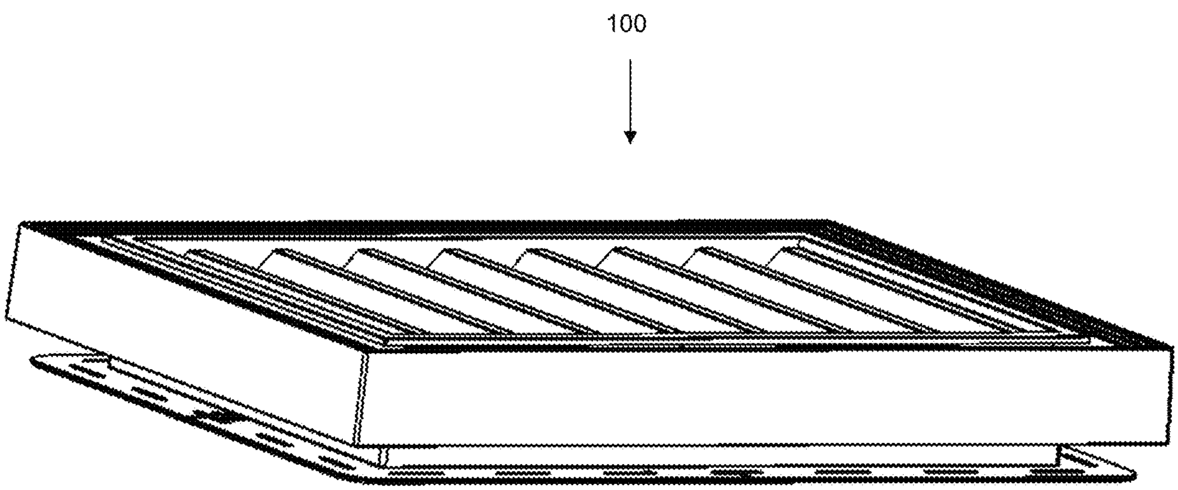


FIG. 7

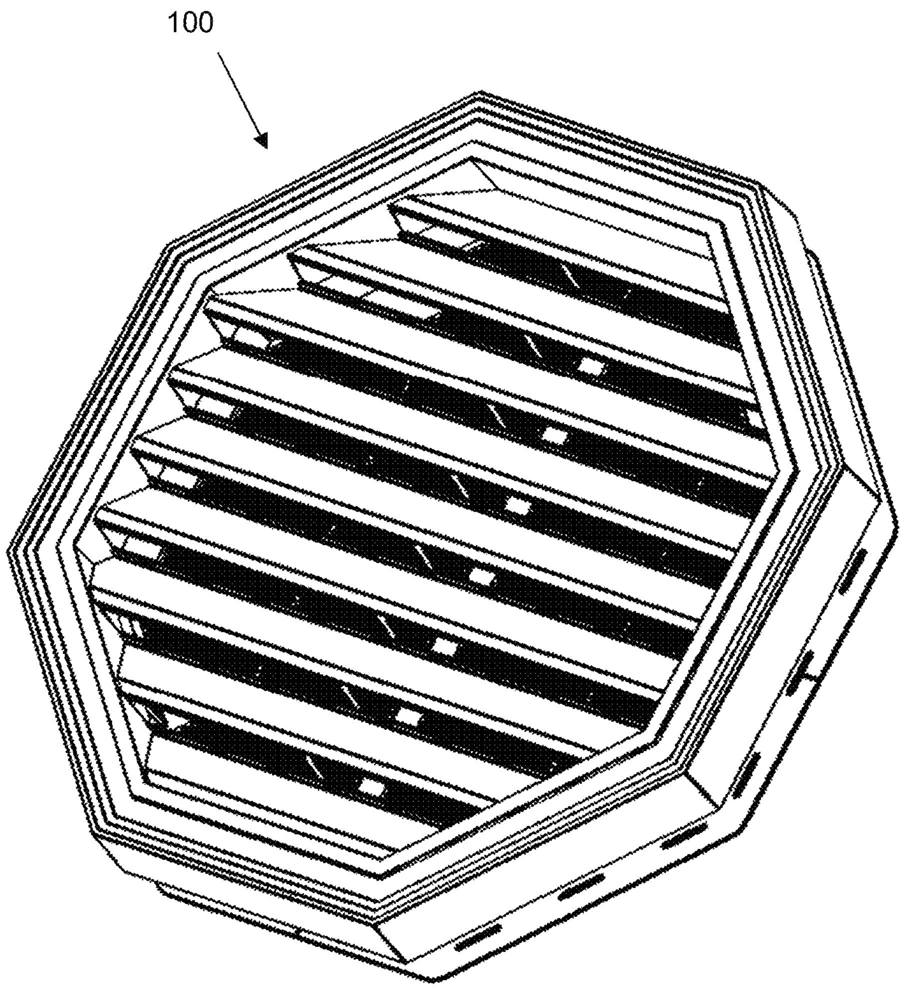


FIG. 8

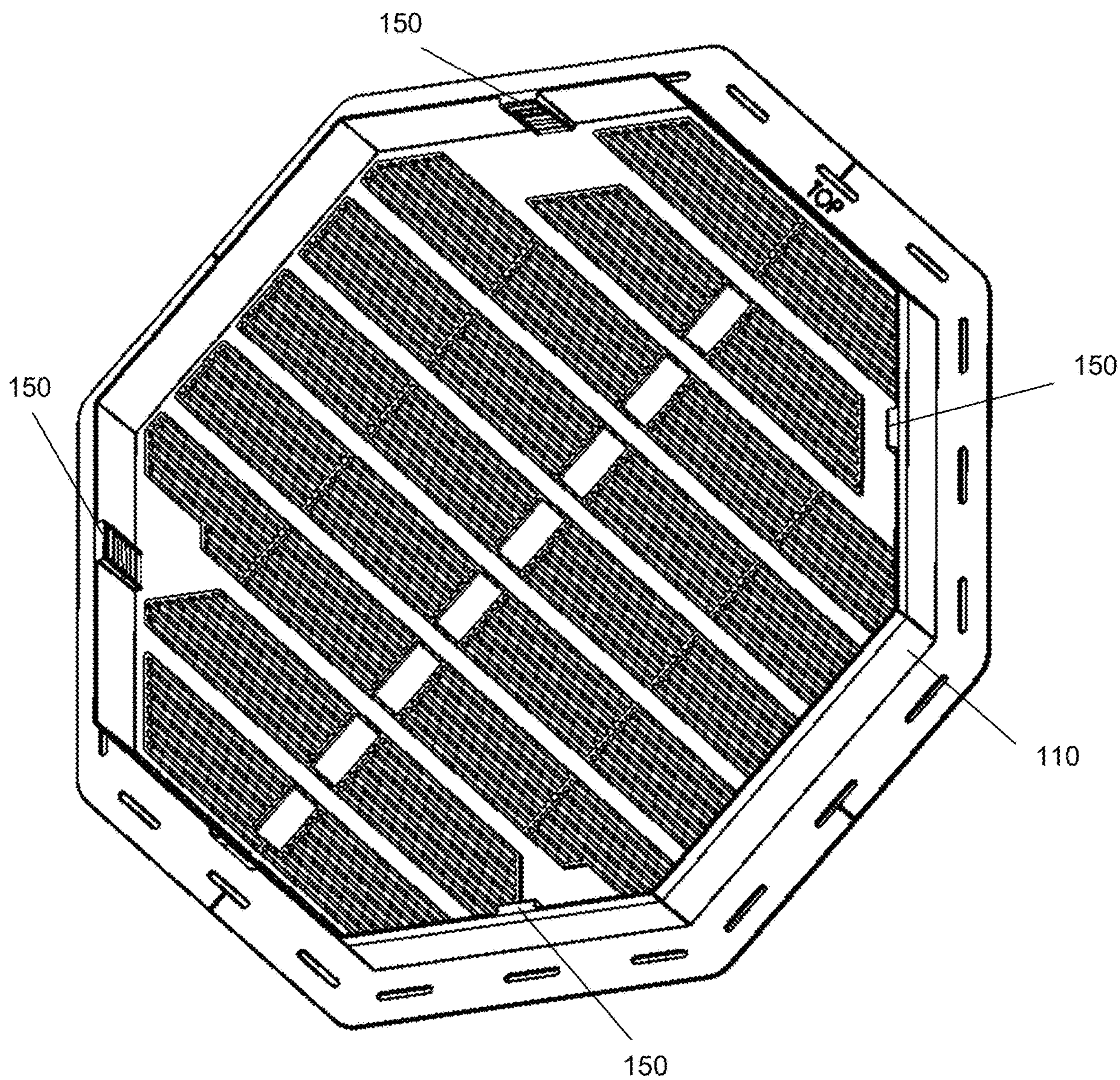


FIG. 9

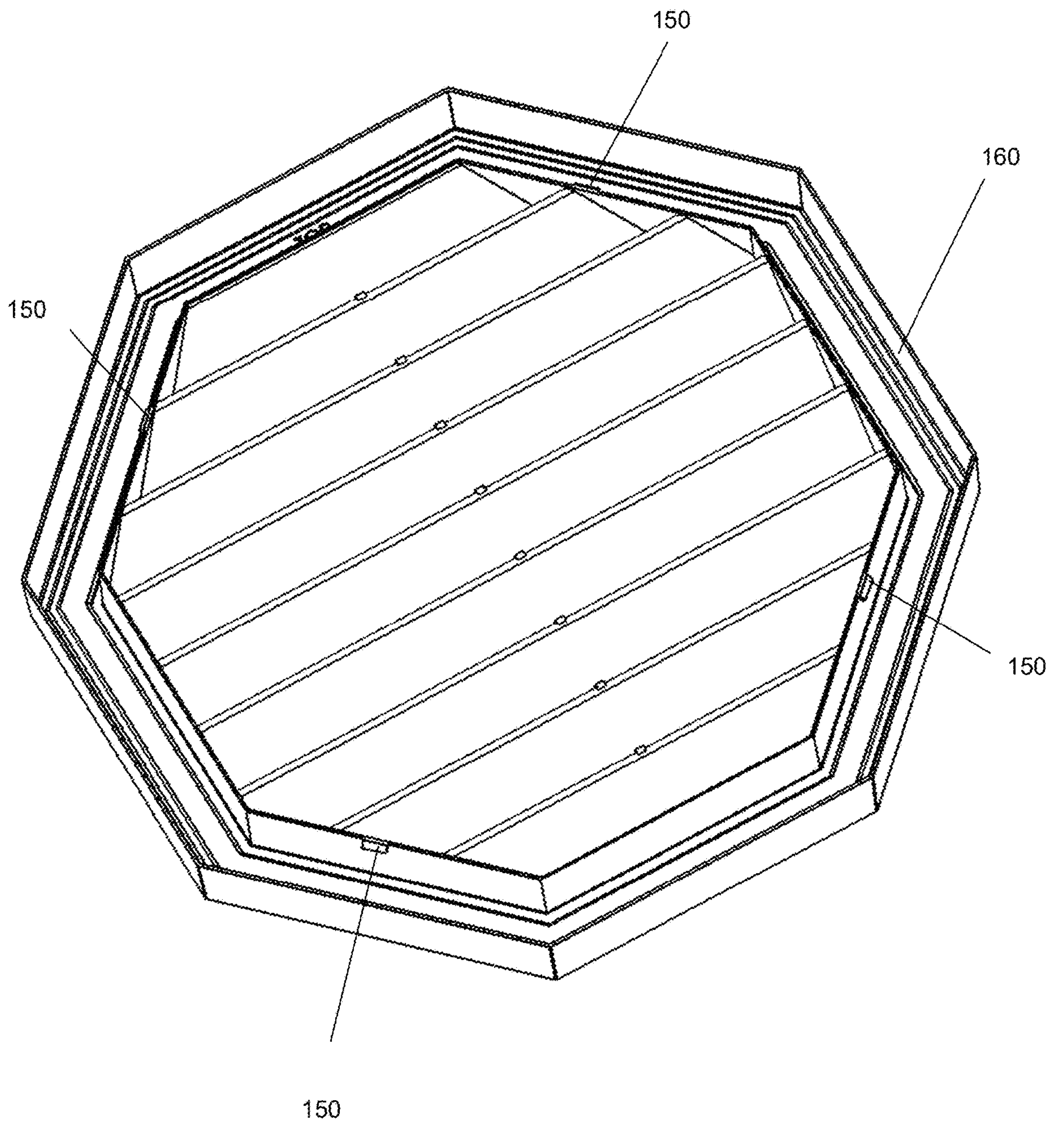


FIG. 10

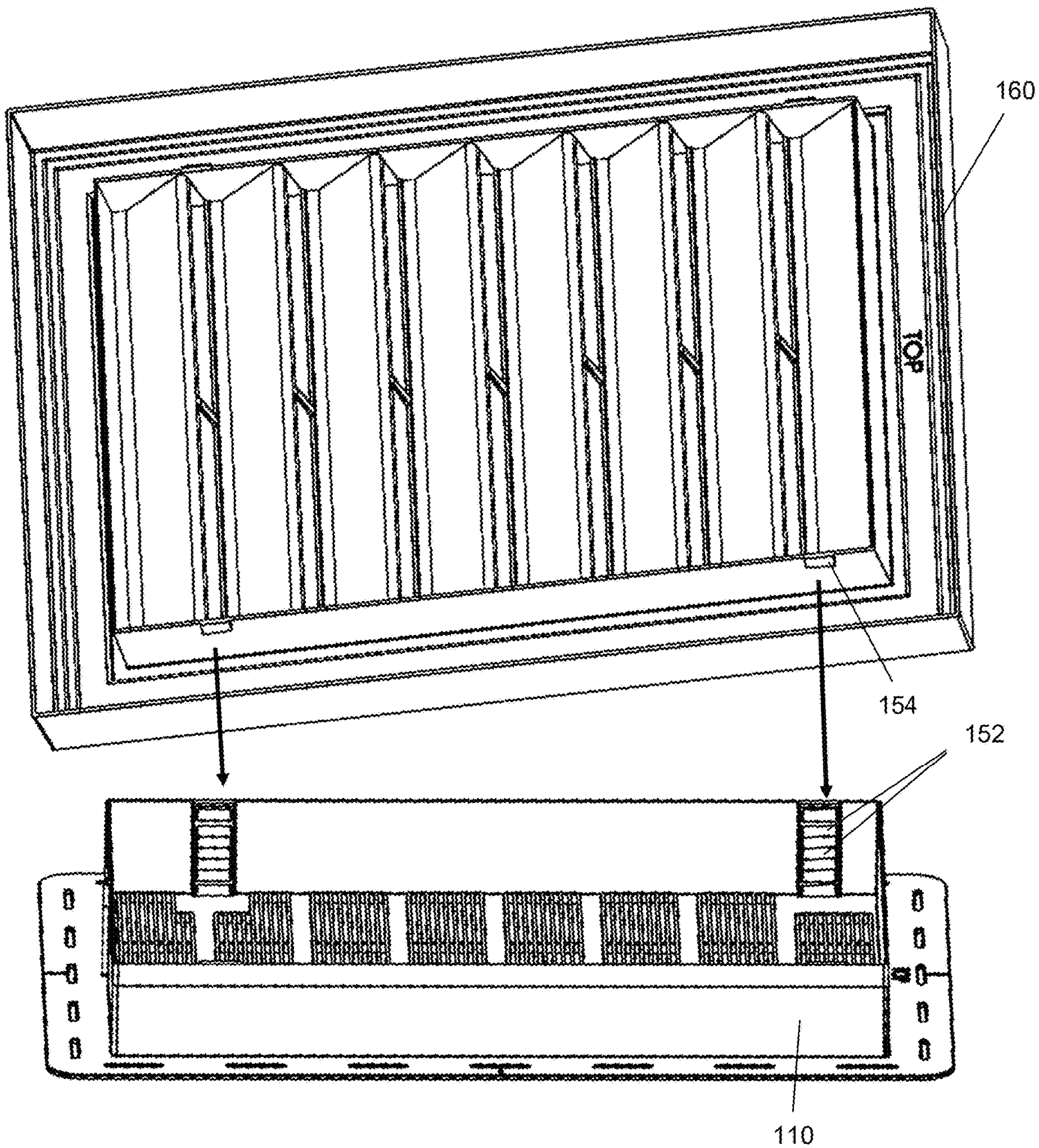


FIG. 11

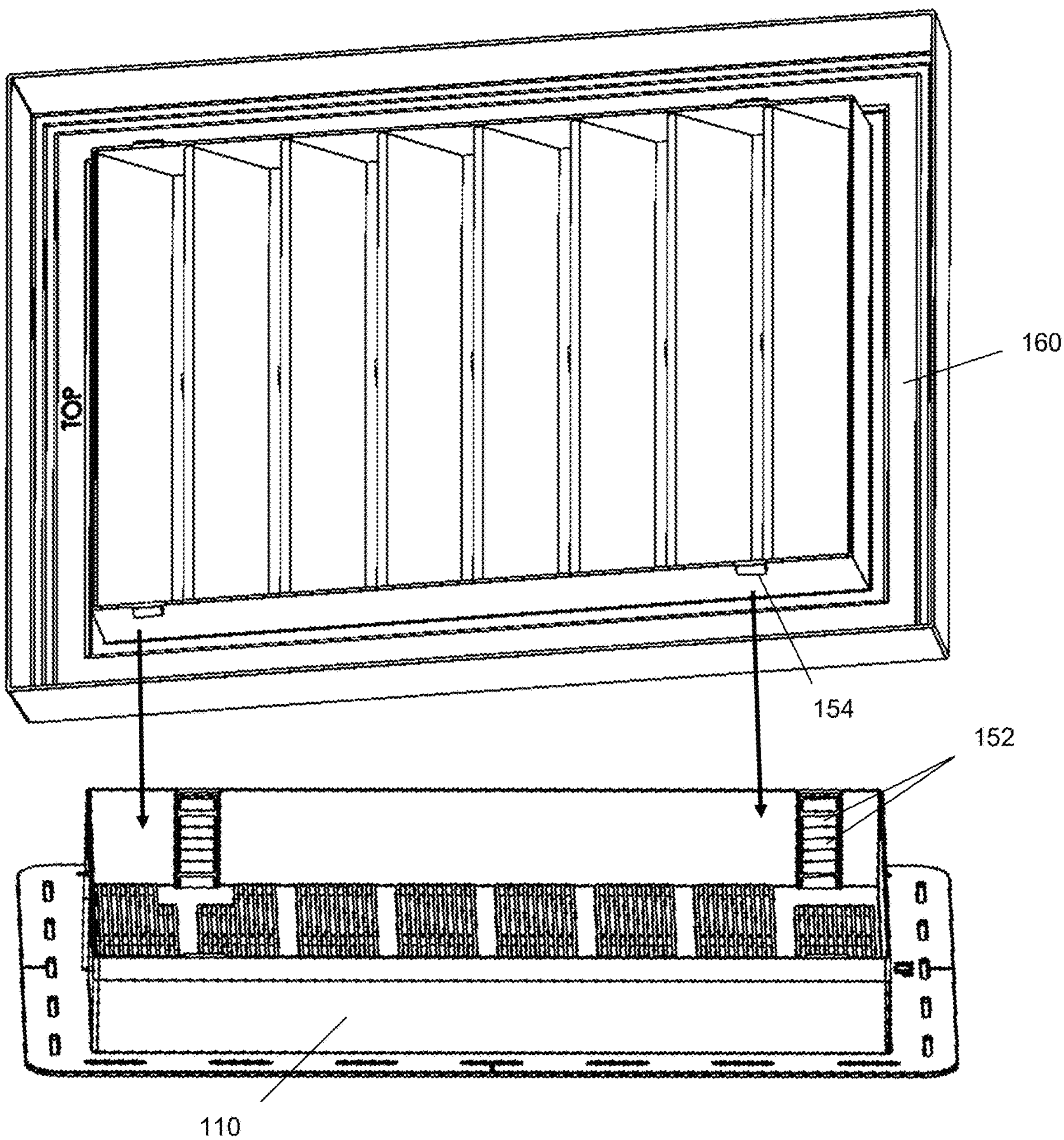


FIG. 12

GABLE VENT ASSEMBLY WITH OFFSET LOCK

BACKGROUND OF THE INVENTION

The present invention relates to airflow devices installed on attic walls, including gable vent assemblies. A gable vent assembly may include a base portion with airflow openings and a cover portion with vents. A cover portion may have a decorative shape, such a square, rectangular, octagonal, and other geometric shapes.

Often storage, transport and shipping of gable assemblies can present a challenge since the two pieces of a base and cover of the assembly can take up considerable space when packed on top of one another. There is therefore a need for a gable assembly that can be packed to take up less space for storage and transport, such as nesting together, while the base and cover can be readily removed from the packed positions and moved into an installation and use position in which the base and cover can be locked to one another when installed in an attic wall. Further, there is a need for a base and cover of a gable assembly to selectively be unlocked when packed or nested and then the base and cover be able to be locked together when installed after being more easily removed from the packed and unlocked configuration..

SUMMARY OF THE INVENTION

To address these needs, in aspects, the technologies described herein relate to a gable vent assembly including: a base including a bottom surface with a plurality of openings; a first base side surface extending outward from the bottom surface, wherein the first base side surface includes a first end of the first base side surface and a second end of the first base side surface, and wherein the first base side surface further includes a first base lock located a first base length from the first end of the first base side surface and a second base lock located a second base length from the second end of the first base side surface, and wherein the first base length has a different measurement from the second base length; a second base side surface opposite the first base side surface extending outward from the bottom surface, wherein the second base side surface includes a first end of the second base side surface and second end of the second base side surface, and wherein the second base side surface further includes a third base lock located a third base length from the first end of the second base side surface and a fourth base lock located a fourth base length from the second end of the second base side surface, and wherein the third base length has a different measurement from the fourth base length; a cover including a plurality of vents between a first cover side surface and a second cover side surface opposite the first cover side surface, wherein the first cover side surface includes a first end of the first cover side surface and a second end of the first cover side surface, and wherein the first cover side surface further includes a first cover lock located a first cover length from the first end of the first cover side surface and a second cover lock located a second cover length from the second end of the first cover side surface, and wherein the first cover length has a different measurement from the second cover length; a second cover side surface opposite the first cover side surface, wherein the second cover side surface includes a first end of the second cover side surface and a second end of the second cover side surface, and wherein the second cover side surface further includes a third cover lock located a third cover length from the first end of the second cover side surface and a fourth cover lock located a fourth cover length from the second end of the second cover side surface, and wherein the third cover length has a different measurement from the fourth cover length; and in a first mating position between the base and the cover, the first base lock couples to the first cover lock, the second base lock couples to second cover lock, the third base lock couples to the third cover lock, and the fourth base lock couples to the fourth cover lock.

In some aspects, the technologies described herein relate to a gable vent assembly, wherein in a second mating position with the cover rotated 180 degrees from the second mating position, the base and cover nest together without any of the locks coupling to one another.

In some aspects, the technologies described herein relate to a gable vent assembly, wherein the locks that couple together are configured to secure together to provide adjustable distance between the bottom surface of the base and the plurality of vents of the cover.

In some aspects, the technologies described herein relate to a gable vent assembly, wherein at least one lock of a pair of locks that couple together include locking teeth.

In some aspects, the technologies described herein relate to a gable vent assembly, wherein at least one lock of a pair of locks that couple together includes a locking projection that engage the locking teeth of the other lock of the pair of locks.

In some aspects, the technologies described herein relate to a gable vent assembly, wherein the cover is a rectangle or square.

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In some aspects, the technologies described herein relate to a gable vent assembly, the cover is a rectangle or square.

In some aspects, the technologies described herein relate to a gable vent assembly, wherein the cover is a rectangle or square.

In some aspects, the technologies described herein relate to a gable vent assembly, wherein the cover is a rectangle or square.

In some aspects, the technologies described herein relate to a gable vent assembly including a base having a bottom surface with openings and the base including a plurality of base locks; and a cover having vents and the cover including a plurality of cover locks, wherein the plurality of base locks align with the plurality of cover locks and securely couple the base to the cover in a first mating position, and wherein the cover is configured to be rotated to a second mating position with the cover and base nesting together without the plurality of base locks coupling to the plurality of cover locks.

In some aspects, the technologies described herein relate to a gable vent assembly wherein the cover and base each include side surfaces in an octagon shape.

In some aspects, the technologies described herein relate to a gable vent assembly wherein the cover and base each include side surfaces in a square shape.

In some aspects, the technologies described herein relate to a gable vent assembly wherein the cover and base each include side surfaces in a rectangle shape.

In some aspects, the technologies described herein relate to a gable vent assembly, wherein the cover is configured to rotate 180 degrees from the first mating position to the second mating position.

In some aspects, the technologies described herein relate to a gable vent assembly, wherein the cover rotates 180 degrees from the first mating position to the second mating position.

In some aspects, the technologies described herein relate to a gable vent assembly, wherein the base locks and cover locks are configured to lock the base and cover at different distances between the bottom surface of the base and the vents of the cover.

In some aspects, the technologies described herein relate to a gable vent assembly, wherein the base locks and cover locks are configured to lock the base and cover at different distances between the bottom surface of the base and the vents of the cover.

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BRIEF DESCRIPTION OF THE DRAWINGS

FIG. **1** is an assembly view from above of a gable vent assembly including a base and cover in one embodiment of the invention.

FIG. **2** is an assembly view from below of a gale vent assembly including a base and cover in one embodiment of the invention.

FIG. **3** is a bottom plan view of a base of a gable vent assembly in one embodiment of the invention.

FIG. **4** i is a top plan view of a base of a gable vent assembly in one embodiment of the invention.

FIG. **5** is a bottom plan view of a cover of a gable vent assembly in one embodiment of the invention.

FIG. **6** is a top plan view of a cover of a gable vent assembly in one embodiment of the invention.

FIG. **7** is a perspective view of a gale vent assembly having a rectangular shape including a base and cover coupled to one another in one embodiment of the invention.

FIG. **8** is a perspective view of a gale vent assembly having an octagonal shape including a base and cover coupled to one another in one embodiment of the invention.

FIG. **9** is a perspective view from above of a top of a base of a gable vent assembly having an octagonal shape in one embodiment of the invention.

FIG. **10** is a perspective view from above of a top of a cover of a gable vent assembly having an octagonal shape in one embodiment of the invention.

FIG. **11** is an assembly view of a gale vent assembly including a base and cover in a first mating position where locks align with one another to couple and lock the assembly together in one embodiment of the invention.

FIG. **12** is an assembly view of a gale vent assembly including a base and cover in a second mating position where locks do not align with one another so that the base and core nest together without locking in one embodiment of the invention.

DETAILED DESCRIPTION OF THE INVENTION

The following elements are described herein with reference to the corresponding reference designations:

ELEMENT	REFERENCE
Gable Vent Assembly	100
Base	110
Base Bottom Surface	115
Bottm Surface Openings	117
First Base Side Surface	122
Second Base Side Surface	124
Base Fastening Flange	126
Fastener Opening	128
First End of the First Base Side Surface	130
Second End of the First Base Side Surface	132
First Base Length	134
Second Base Length	136
First End of Second Base Side Surface	140
Second End of Second Base Side Surface	142
Third Base Length	144
Fourth Base Length	146
Locks	150
First Base Lock	150a
Second Base Lock	150b
Third Base Lock	150c
Fourth Base Lock	150d
First Cover Lock	150e
Second Cover Lock	150f
Third Cover Lock	150g
Fourth Cover Lock	150h
Locking Teeth	152
Locking Projection	154
Cover	160
Cover Vents	165
First Cover Side Surface	172
Second Cover Side Surface	174
First End of First Cover Side Surface	180
Second End of First Cover Side Surface	182
First Cover Length	184
Second Cover Length	186
First End of Second Cover Side Surface	190
Second End of Second Cover Side Surface	192
Third Cover Length	194
Fourth Cover Length	196

Referring to FIGS. **1** and **2**, a gable vent assembly **100** includes a cover **160** with vents **165** that couples to a base **110**. The cover **160** may include numerous decorative shapes, such as a square (not shown), rectangle (shown in FIGS. **1-7**, **11** and **12**), octagon (shown in FIGS. **8-10**), other polygons, circles, rounded and oval shapes, and a variety of other symmetrical and non-symmetrical geometries. The base **110** can have a similar or nearly similar shape to the cover **160** or can have a different shape in some embodiments that still permit coupling of the base **110** and cover **160**, while the cover **160** only has the desired shape since the base **110** is hidden when installed. A decorative cover **160** and vents **165** is secured to the base **110** facing outward on an exterior of an attic wall. The base **110**, such as including a fastening flange **126** with fastener openings **128** as shown in FIG. **3**, is fastened to the interior of an attic wall so that airflow can pass though the gable vent assembly between the cover **160** at the exterior of the attic wall and then through openings **117** in a bottom surface **115** of the base that faces into the interior of an attic.

With further reference to FIGS. **3** and **4**, the base **110** includes a bottom surface **115** with a plurality of openings **117**. A first base side surface **122** extends outward from the bottom surface **115**. The first base side surface **122** includes a first end **130** of the first base side surface and a second end **132** of the first base side surface, The first base side surface **122** further includes a first base lock **150***a* located a first base length **134** from the first end **130** of the first base side surface and a second base lock **150***b* located a second base length **136** from the second end **132** of the first base side surface, To permit rotation or other movement of the cover and base from a mating position of nesting in an unlocked position during packaging to another mating position of being coupled and locked together when installed in an attic wall, the first base length **134** has a different measurement from the second base length **136**.

The base **110** also includes a second base side surface **124** opposite the first base side surface **122** extending outward from the bottom surface **115**, The second base side surface **124** includes a first end **140** of the second base side surface and second end **142** of the second base side surface. The second base side surface **124** further includes a third base lock **150***c* located a third base length **144** from the first end **140** of the second base side surface and a fourth base lock **150***d* located a fourth base length **146** from the second end **142** of the second base side surface. To permit rotation or other movement of the cover and base from a mating position of nesting in an unlocked position during packaging to another mating position of being coupled and locked together when installed in an attic wall, the third base length **144** has a different measurement from the fourth base length **146**.

Referring to FIGS. **5** and **6**, a cover **160** includes a plurality of vents **165** between a first cover side surface **172** and a second cover side surface **174** opposite the first cover side surface.

The first cover side surface **172** includes a first end **180** of the first cover side surface and a second end **182** of the first cover side surface. The first cover side surface **172** further includes a first cover lock **150***e* located a first cover length **184** from the first end **180** of the first cover side surface and a second cover lock **150***f* located a second cover length **186** from the second end **182** of the first cover side surface. The first cover length **184** has a different measurement from the second cover length **186** to permit rotation or other movement of the cover **160** and base **110** from a mating position of nesting in an unlocked position during packaging, when the locks **150** of the base **110** and cover **160** do not align with one another as shown in FIG. **12**, to another mating position of being coupled and locked together during installation, when the locks **150** of the base **110** and cover **160** align with one another as shown in FIG. **11**.

The cover **160** includes a second cover side surface **174** opposite the first cover side surface **172**. The second cover side surface **174** includes a first end **190** of the second cover side surface and a second end **192** of the second cover side surface, The second cover side surface **174** further includes a third cover lock **150***g* located a third cover length **194** from the first end **190** of the second cover side surface and a fourth cover lock **150***h* located a fourth cover length **196** from the second end **192** of the second cover side surface. The third cover length **194** has a different measurement from the fourth cover length **196** to permit rotation or other movement of the cover **160** and base **110** from a mating position of nesting in an unlocked position during packaging, when the locks **150** of the base **110** and cover **160** do not align with one another as shown in FIG. **12**, to another mating position of being coupled and locked together during installation, when the locks **150** of the base **110** and cover **160** align with one another as shown in FIG. **11**.

Referring to FIG. **11**, a first mating position between the base **110** and the cover **160** is shown wherein the first base lock couples to the first cover lock, the second base lock couples to second cover lock, the third base lock couples to the third cover lock, and the fourth base lock couples to the fourth cover lock since all of the locks **150** align with one another in the first mating position. In some embodiments, a lock **150** of either the base **110** or the cover **160** may include a plurality of teeth **152** in a row and spaced apart at desired distance that interface with a complementary locking projection **154** so that the cover and base may be distance-adjusted with respect to one another. The ability of the locks to accommodate adjustment of the cover either toward or away from the bottom surface **115** of the base **110** enables the gable assembly to be installed more easily through attic wall having different widths between the interior attic wall that couples to the base **110** and the exterior attic wall where the cover **160** is positioned and couples to the base. In other embodiments of the invention, sliding mechanisms, push button spring clip and hole connections, clips, hook and loop material, spring connectors and like locking mechanisms provide alternative coupling mechanisms for teeth and a locking projection.

Referring to FIG. **12**, a second mating position is shown wherein the locks **150** of the base **110** and cover **160** do not align and permit the base and cover to be nested together without locking. In one embodiment, the cover **160**, while not locked and nested with the base **110**, can be rotated 180 degrees from the first mating position shown in FIG. **11** where the locks **150** of the cover **160** and base **110** would align with one another to the second mating position where the locks **150** do not align with another and can advantageously nest, and take up less room, in an unlocked position for storage, transport and shipping until the gable assembly needs to be assembled. When installation of the gable vent assembly is desired, the unlocked base **110** and cover **160** can be easily separated from the nesting position and then rotated to align the locks to couple and lock together the base **110** and cover **160**. As described, in some embodiment the locks include distance selection capability between the cover **160** and the bottom surface **115** of the base **110**, such as with locking teeth **152** (shown on the base **110**, but could alternatively be included on the cover **160**) and a locking projection **154** (shown on the cover **160**, but could alternatively be included on the base **110**).

Referring to FIGS. **8-10**, an alternative embodiment of a gable vent assembly **100** is shown having an octagonal shape. In the depicted embodiment, multiple side surfaces of both the base **110** and cover **160** may each include at least one lock **150**. It will be appreciated that rotation of the cover **160** or base **110**, wherein both have an octagonal shape, permit a base **110** to mate with a cover **160** in both locked and unlocked nesting positions, since the locks **150** of base **110** (FIG. **9**) can be aligned with the locks **150** of cover **160** (FIG. **10**) for coupling and locking or the base **110** or cover **160** can be rotated so that a surface of a cover **160** with a lock will abut a surface of the base **110** without a lock and allow nesting without alignment an locking together of the locks **150**.

Various embodiments of the invention have been described. It will, however, be evident that various modifications and changes may be made thereto, and additional embodiments may be implemented, without departing from the broader scope of the invention disclosed herein. This specification is to be regarded in an illustrative rather than a restrictive sense.

What is claimed is:

1. A gable vent assembly comprising:

a base including

a bottom surface with a plurality of openings;

a first base side surface extending outward from the bottom surface, wherein the first base side surface includes a first end of the first base side surface and a second end of the first base side surface, and wherein the first base side surface further includes a first base lock located a first base length from the first end of the first base side surface and a second base lock located a second base length from the second end of the first base side surface, and wherein the first base length has a different measurement from the second base length;

a second base side surface opposite the first base side surface extending outward from the bottom surface, wherein the second base side surface includes a first end of the second base side surface and second end of the second base side surface and wherein the second base side surface further includes a third base lock located a third base length from the first end of the second base side surface and a fourth base lock located a fourth base length from the second end of the second base side surface, and wherein the third base length has a different measurement from the fourth base length;

a cover including a plurality of vents between a first cover side surface and a second cover side surface opposite the first cover side surface, wherein

the first cover side surface includes a first end of the first cover side surface and a second end of the first cover side surface, and wherein the first cover side surface further includes a first cover lock located a first cover length from the first end of the first cover side surface and a second cover lock located a second cover length from the second end of the first cover side surface, and wherein the first cover length has a different measurement from the second cover length;

a second cover side surface opposite the first cover side surface, wherein the second cover side surface includes a first end of the second cover side surface and a second end of the second cover side surface, and wherein the second cover side surface further includes a third cover lock located a third cover length from the first end of the second cover side surface and a fourth cover lock located a fourth cover length from the second end of the second cover side surface, and wherein the third cover length has a different measurement from the fourth cover length;

in a first mating position between the base and the cover, the first base lock couples to the first cover lock, the second base lock couples to second cover lock, the third base lock couples to the third cover lock, and the fourth base lock couples to the fourth cover lock; and

wherein in a second mating position with the cover rotated 180 degrees from the first mating position, the base and the cover nest together without any of the locks coupling to one another.

2. The gable vent assembly of claim **1**, wherein the locks that couple together are configured to secure together to provide adjustable distance between the bottom surface of the base and the plurality of vents of the cover.

3. The gable vent assembly of claim **2**, wherein at least one lock of a pair of locks that couple together include locking teeth.

4. The gable vent assembly of claim **3**, wherein at least one lock of a pair of locks that couple together includes a locking projection that engage the locking teeth of the other lock of the pair of locks.

5. The gable vent assembly of claim **4**, wherein the cover is a rectangle or square.

6. The gable vent assembly of claim **3**, wherein the cover is a rectangle or square.

7. The gable vent assembly of claim **2**, wherein the cover is a rectangle or square.

8. The gable vent assembly of claim **1**, wherein the cover is a rectangle or square.

* * * * *